TokenRouter: Uncertainty-Guided Token-Level Routing for Efficient LLM Reasoning

Anonymous ACL submission

Abstract

Large Language Models (LLMs) are strong at multi-step reasoning, but running them for every generated token is expensive. Small language models (SLMs) are much cheaper, yet less reliable at the hardest reasoning steps. Existing routing methods mostly make a single query-level decision, assuming that response difficulty is uniform once decoding begins. For chain-of-thought reasoning, however, difficulty varies during generation, making token-level routing a more appropriate granularity. This setting introduces three challenges: token-level uncertainty is noisy, local routing decisions must satisfy global efficiency constraints, and frequent switching can disrupt reasoning coherence. We present TokenRouter, a black-box runtime controller for uncertainty-guided token-level routing between small and large language models. TokenRouter combines a robust entropy detector, a self-calibrating control policy, and a dynamic intervention commitment mechanism for stable switching over difficult spans. Across three reasoning benchmarks, TokenRouter maintains over 99% of the reasoning accuracy of a pure LLM while activating the LLM for only 56.2% to 76.0% of generated tokens. Code is available at https://anonymous.4open.science/r/TokenRouter.

1 Introduction

Large Language Models (LLMs) have become strong reasoners on multi-step tasks, but this capability is expensive to deploy because every generated token requires a full forward pass through a large model (Kojima et al., 2022; Wei et al., 2022; Yao et al., 2023; Liu et al., 2025). This cost is especially pronounced for chain-of-thought reasoning, where solving a single problem may require long autoregressive traces. Small language models (SLMs) offer a much cheaper alternative, but relying on them throughout decoding can hurt accuracy at the most difficult reasoning steps. This creates a practical tension in efficient LLM reasoning: always using the large model wastes computation, while always using the small model risks failure at the most critical moments.

Model routing is a natural response to this trade-off. Prior work typically routes at the query level: a router examines the input once and then assigns the entire request to a single model (Hu et al., 2024; Ong et al., 2024; Tran et al., 2025). This design is effective when difficulty is largely determined by the input as a whole, but reasoning traces are rarely uniform. Within a single response, easy procedural spans and difficult reasoning steps often alternate (Yu et al., 2025; Kim et al., 2025). As a result, query-level routing uses the wrong granularity for chain-of-thought reasoning. It cannot selectively allocate stronger model capacity to the parts of the generation that actually need it.

Our starting point is therefore to treat efficient reasoning as a runtime collaboration problem between small and large language models. The key question is not only which model should answer a query, but also when the system should escalate from the SLM to the LLM during generation, and for how long. This perspective motivates token-level routing. It also distinguishes our setting from one-shot routing and from training-heavy token routing schemes: we seek a black-box test-time controller that can operate on unmodified models using signals that are available at decoding time.

Making token-level routing work in practice is challenging. First, token-level uncertainty signals are noisy, so naive switching based on raw scores can trigger spurious interventions (Raposo et al., 2024; Wu et al., 2025). Second, local routing decisions must still satisfy a global efficiency target rather than overusing the LLM whenever uncertainty spikes. Third, per-token switching can oscillate and degrade the overall quality of reasoning (Pan et al., 2025; Zheng et al., 2025). A useful

routing mechanism must therefore be uncertainty-guided, budget-aware, and stability-aware at the same time.

To address these challenges, we introduce TokenRouter, a closed-loop framework for uncertainty-guided token-level routing. TokenRouter uses the SLM's output entropy as a deployable uncertainty signal for black-box runtime control. A robust entropy detector first smooths the raw entropy stream into a stable estimate of local reasoning difficulty, making the routing signal less sensitive to token-level noise. A self-calibrating control policy then converts a target usage budget into an adaptive switching threshold, allowing the system to regulate LLM usage online rather than relying on a fixed trigger. Finally, a dynamic intervention commitment mechanism keeps the LLM active long enough to cover difficult spans, reducing oscillation and preserving reasoning coherence. Together, these components form a unified runtime policy for stable small-large collaboration during decoding.

We evaluate TokenRouter on three reasoning benchmarks spanning different difficulty levels (Cobbe et al., 2021; Hendrycks et al., 2021; Li et al., 2024). Across these benchmarks, TokenRouter maintains over 99% of the reasoning accuracy of a powerful LLM (e.g., DeepSeek-R1-Distill-Qwen-14B) while activating it for only 56.2% to 76.0% of the generated tokens, with the remaining tokens handled by a much smaller model (e.g., DeepSeek-R1-Distill-Qwen-1.5B). These results show that uncertainty-guided small-large collaboration can achieve a strong accuracy-efficiency trade-off for reasoning without modifying the two underlying models.

Our contributions are as follows:

- We formulate efficient reasoning as an uncertainty-guided token-level routing problem between small and large language models, targeting the granularity mismatch of query-level routing for chain-of-thought generation.
- We propose TokenRouter, a black-box closed-loop runtime controller that integrates robust uncertainty estimation, self-calibrated efficiency control, and stability-aware intervention during decoding.
- We present a deployment-oriented implementation that keeps routing lightweight through CPU-side decisions, KV-cache reuse, and minimal additional memory overhead.
- We show on multiple reasoning benchmarks that TokenRouter delivers a strong accuracy-efficiency trade-off, retaining over 99% of pure-LLM accuracy while reducing large-model usage to 56.2%–76.0% of tokens.

2 Related Work

2.1 Efficient LLM Inference

To reduce the deployment cost of LLMs, prior work has explored model compression and dynamic computation. Most of these methods are white-box, requiring access to model internals. Static compression techniques (Wang et al., 2024; Fang et al., 2024), including quantization (Xiao et al., 2023), pruning (Sun et al., 2023), and knowledge distillation (Gu et al., 2023), permanently shrink models but typically require retraining or fine-tuning. Dynamic methods such as early exiting (Bae et al., 2023; Chen et al., 2023) adapt computation at runtime, yet remain tightly coupled to specific architectures and do not naturally support collaboration across independent models. In contrast, our method is a black-box framework that coordinates unmodified models at inference time and is therefore complementary to these approaches.

2.2 Collaborative Computing Architectures

Combining multiple computational units is a common strategy for balancing performance and efficiency, including deployment settings that coordinate on-device small models with cloud-based large models or other heterogeneous inference resources (Chen et al., 2025; Niu et al., 2025). The Mixture-of-Experts (MoE) architecture (Dai et al., 2024; Yun et al., 2024) scales model capacity through sparse expert activation within a single system, while speculative decoding (Xia et al., 2024) pairs a small draft model with a large verifier to reduce latency. These methods coordinate computation in different ways, but they are typically designed for acceleration rather than selective capacity allocation over a reasoning trace. Our goal is different: we dynamically schedule between independent models based on real-time reasoning difficulty to reduce overall cost while preserving reasoning quality.

2.3 Model Routing

Model routing directs an input request to the most suitable model within a model suite (Jitkrittum et al., 2025; Varangot-Reille et al., 2025). Most

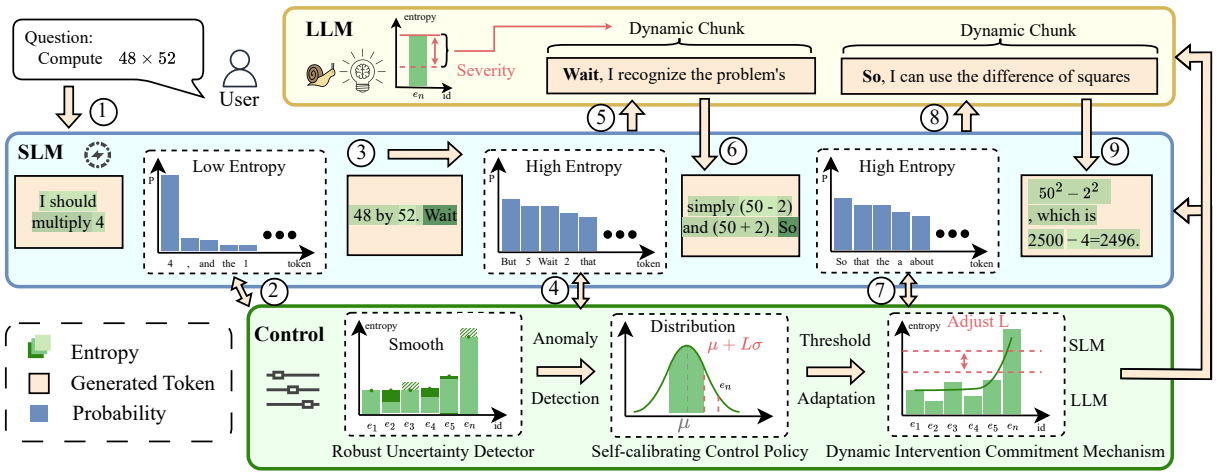


Figure 1: An overview of the TokenRouter framework, where numbered steps illustrate the adaptive token-level routing process for a sample reasoning task. Its control module continuously monitors the SLM's output entropy, triggering an intervention from the more capable LLM only at moments of high local reasoning difficulty to ensure both efficiency and accuracy.

existing methods are query-level (Hu et al., 2024): a classifier assesses the query once and routes it to a single model (Ong et al., 2024; Tran et al., 2025). Because this decision is fixed before generation, it cannot adapt to complexity that changes within a response. Some recent approaches also consider lightweight or confidence-based routing, but they still largely assume that the routing unit is the whole request. Our work instead performs token-level routing, re-evaluating decisions during decoding to enable finer-grained allocation. This dynamic policy can also complement an initial query-level routing decision when a coarse decision is still useful.

3 Methodology

We propose TokenRouter, an adaptive runtime framework for efficient LLM reasoning via token-level routing. As shown in Figure 1, the framework combines three components: a robust entropy detector that turns noisy token entropy into a stable estimate of local reasoning difficulty, a self-calibrating control policy that converts a target compute budget into adaptive activation thresholds, and a dynamic intervention commitment mechanism that prevents oscillatory switching and preserves coherence during difficult spans.

3.1 Problem Formulation

We formulate adaptive inference as a constrained sequential decision process that dynamically selects between a large language model (LLM) and a small language model (SLM). At each timestep t, the policy observes the current prefix state $s_t = (y_1, \ldots, y_{t-1})$, chooses an action $a_t \in \{\text{LLM}, \text{SLM}\}$, generates the next token $y_t \sim p_{a_t}(\cdot|s_t)$, and transitions to s_{t+1}. The objective is to find a policy π^* that maximizes the expected utility $U(Y)$ of the final sequence $Y = (y_1, \ldots, y_T)$ subject to a global budget on the LLM usage rate R_{target}:

$$\pi^* = \arg\max_{\pi} \mathbb{E}_{Y\sim\pi}[U(Y)] \tag{1}$$

$$\text{subject to} \quad \frac{1}{T}\sum_{t=1}^{T} \mathbb{I}(\pi(s_t) = \text{LLM}) \le R_{\text{target}} \tag{2}$$

where $\mathbb{I}(\cdot)$ is the indicator function and $U(Y)$ is an external metric, such as final-answer accuracy. The key challenge is partial observability: the reasoning complexity required at each state s_t is latent and must be inferred online from noisy proxy signals.

3.2 Robust Entropy Detector

The first component addresses the perception challenge: obtaining a reliable real-time estimate of local reasoning difficulty. We use the Shannon entropy of the model's output distribution P_t as an observable proxy. Here, black-box means that the controller does not access hidden states, modify model parameters, or require additional supervised router training, and only uses next-token probabilities available at decoding time. The raw entropy at

timestep t, denoted as z_t, is:

$$z_t = H(P_t) = -\sum_{v \in V} p_t(v) \log p_t(v) \quad (3)$$

The raw signal is highly volatile because it conflates genuine shifts in reasoning complexity with stochastic noise from autoregressive generation and local word variation. As a result, raw entropy alone is an unreliable basis for stable control.

We therefore model the underlying reasoning difficulty x_t as a latent variable in a state-space model. Assuming that this difficulty evolves smoothly over time, we use the following state evolution equation:

$$x_t = Fx_{t-1} + \epsilon_{t-1}, \quad \epsilon_{t-1} \sim \mathcal{N}(0, Q) \quad (4)$$

The observation equation links this latent state to the noisy entropy measurement:

$$z_t = Hx_t + \nu_t, \quad \nu_t \sim \mathcal{N}(0, R) \quad (5)$$

We set $F = 1$, reducing the latent dynamics to a random walk, and $H = 1$, treating the observed entropy z_t as a direct linear measurement of the underlying difficulty x_t. The process noise ϵ_{t-1} captures unpredictable shifts in reasoning difficulty, while the measurement noise ν_t accounts for stochasticity in the entropy signal.

This formulation yields an efficient closed-form estimate of the posterior distribution $p(x_t|z_{1:t})$ through a standard predict-update cycle. The prediction step forecasts the current prior:

$$\hat{x}_{t|t-1} = F\hat{x}_{t-1|t-1} \quad (6)$$

$$P_{t|t-1} = FP_{t-1|t-1}F^T + Q \quad (7)$$

The update step then incorporates the new observation z_t:

$$K_t = P_{t|t-1}H^T(HP_{t|t-1}H^T + R)^{-1} \quad (8)$$

$$\hat{x}_{t|t} = \hat{x}_{t|t-1} + K_t(z_t - H\hat{x}_{t|t-1}) \quad (9)$$

$$P_{t|t} = (I - K_tH)P_{t|t-1} \quad (10)$$

The resulting posterior mean $\hat{x}_{t|t} = \mathbb{E}[x_t|z_{1:t}]$ provides a smooth estimate of latent reasoning difficulty and serves as the uncertainty signal for the subsequent control policy.

3.3 Self-Calibrating Control Policy

Having established a robust estimator for latent reasoning difficulty $\hat{x}_{t|t}$, we next address when to trigger LLM intervention while respecting a global compute budget. The main difficulty is that a fixed activation threshold does not generalize well across heterogeneous and non-stationary problem instances, leading to inconsistent performance and less predictable cost.

To address this, we introduce a dynamic self-calibrating policy that detects deviations from an evolving difficulty baseline rather than reacting to absolute complexity. We first transform the filtered estimate $\hat{x}_{t|t}$ into a standardized anomaly score so that routing decisions remain comparable across domains. To do so, we maintain a moving reference for typical reasoning difficulty with an Exponentially Weighted Moving Average (EWMA), updating the signal mean μ_t and variance σ_t^2 as:

$$\mu_t = \lambda\hat{x}_{t|t} + (1 - \lambda)\mu_{t-1} \quad (11)$$

$$\sigma_t^2 = \lambda(\hat{x}_{t|t} - \mu_t)^2 + (1 - \lambda)\sigma_{t-1}^2 \quad (12)$$

where $\lambda \in (0, 1]$ is the smoothing factor. We then compute the standardized anomaly score S_t:

$$S_t = \frac{\hat{x}_{t|t} - \mu_t}{\sigma_t} \quad (13)$$

This score measures how strongly the current state deviates from its recent history and provides a normalized basis for intervention decisions.

The standardized score is the input to a self-calibrating trigger that determines the intervention threshold L_t at each timestep. We engage the LLM if $S_t > L_t$, and update L_t through a proportional feedback controller based on the current usage rate R_t and target budget R^*:

$$L_t = \text{clip}\left(L_{t-1} + \beta \cdot (R_t - R^*), L_{\min}, L_{\max}\right) \quad (14)$$

This formulation creates an adaptive closed-loop system in which $(R_t - R^*)$ acts as the error signal and β controls the trade-off between convergence speed and threshold stability. The controller therefore adapts online to the observed trajectory instead of relying on a preset intervention schedule.

If the system is over-budget ($R_t > R^*$), the threshold rises and future LLM activations become less likely. If it is under-budget, the threshold falls. In this way, the global budget is translated into an adaptive online decision rule without manual threshold tuning.

3.4 Dynamic Intervention Commitment

Having addressed when to intervene, we now address reasoning oscillation. A purely reactive policy that makes an independent routing decision at

each timestep can alternate rapidly between the LLM and SLM, fragmenting the chain of thought and degrading reasoning quality.

Our core insight is that intervention duration should scale with the severity of the anomaly that triggered it. Minor fluctuations may require only brief LLM assistance, whereas sharp spikes often mark the start of a difficult reasoning phase that benefits from sustained intervention. This turns routing from isolated token decisions into span-level support for contiguous hard segments.

Motivated by this insight, we design the dynamic intervention commitment mechanism. When an anomaly at timestep t^* triggers an intervention with a standardized score S_{t^*} (i.e., $S_{t^*} > L_{t^*}$), the system commits to engaging the LLM for at least k subsequent tokens. This commitment duration k is dynamically determined by the following rule:

$$k = \lceil k_{\text{base}} + \alpha \cdot \max(0, S_{t^*} - L_{t^*}) \rceil \tag{15}$$

To further formalize stability, we define a commitment decay function that gradually releases LLM engagement once the observed complexity stabilizes. Let D_t denote the residual commitment strength at step t:

$$D_t = D_{t-1} \cdot e^{-\gamma \Delta t} + \mathbb{I}(S_t > L_t) \tag{16}$$

where γ controls the decay rate and $\mathbb{I}(\cdot)$ is the indicator function for triggered anomalies. This formulation provides a smooth transition between intervention states and avoids premature disengagement during difficult spans.

Finally, to ensure continuity of reasoning, the effective routing decision at each timestep incorporates both the current anomaly signal and the ongoing commitment:

$$a_t = \begin{cases} \text{LLM}, & \text{if } D_t > \tau \\ \text{SLM}, & \text{otherwise} \end{cases} \tag{17}$$

This mechanism mitigates oscillation and preserves chain-of-thought coherence by adapting intervention duration to the severity of the reasoning challenge. It applies stronger stabilization to major anomalies and only brief intervention to minor ones, safeguarding solution quality while maintaining efficiency. In effect, the mechanism introduces temporal hysteresis that keeps the model assignment stable until the reasoning state has stabilized over subsequent reasoning steps.

4 Experiments

4.1 Experimental Setup

Datasets. We evaluate TokenRouter on three reasoning datasets that collectively span a wide spectrum of problem difficulty. The first is *GSM8K* (Cobbe et al., 2021), a dataset of grade school problems requiring multi-step arithmetic. The second is *MATH500* (Hendrycks et al., 2021), a benchmark of high school competition-level problems requiring symbolic manipulation and algebraic reasoning. The third is *AIME24* (Li et al., 2024), which contains challenging problems from the American Invitational Mathematics Examination, emphasizing creative problem-solving and advanced logical deduction. Together, these datasets provide a comprehensive evaluation landscape, covering diverse complexities from basic reasoning to sophisticated multi-hop inference.

Baselines. We compare TokenRouter against several baselines: Pure SLM and Pure LLM establish performance bounds using exclusively small or large models. Uniform Routing applies naive stochastic selection with fixed LLM probability matching our target usage ratio. We include CITER (Zheng et al., 2025), a confidence-based token-level method, and RouteLLM (Ong et al., 2024), which routes entire queries to a single model via its BERT (Devlin et al., 2019) or CASUAL (Wei et al., 2021) classifiers. These baselines comprehensively evaluate routing strategies across different granularities and allocation mechanisms.

Model and Implementation Details. We evaluate TokenRouter using the DeepSeek-R1-Distill-Qwen (Guo et al., 2025) family of models. Unless otherwise specified, we use the 1.5B model as the SLM and the 14B model as the LLM. We set target ratios R_{target} to 0.7 for GSM8K and 0.8 for MATH500 and AIME24. The entropy detector uses $Q = 0.1$, $R = 0.5$. The control policy employs smoothing $\lambda = 0.05$, gain $\beta = 1.0$, initial threshold $L_0 = 2.0$ clipped to $[1.0, 4.0]$. The commitment mechanism sets $\alpha = 2.0$ and k_{base} of 20 for GSM8K and 50 for MATH500 and AIME24. All experiments use a temperature of 0.7 sampling on a server with 80GB NVIDIA H100 GPUs. The routing controller runs on CPU, both models retain KV caches in GPU memory for reuse after switching, entropy is computed from existing output logits without extra GPU memory, and maintaining controller state adds under 10MB of extra CPU memory during runtime.

Table 1: Comparison of TokenRouter against baselines across datasets using two different SLM-LLM combinations. TokenRouter consistently achieves performance comparable to the pure LLM while significantly reducing the LLM usage ratio.

Method	Combination: 1.5B & 7B				Combination: 1.5B & 14B			
	Acc (%) ↑	Average Tokens ↓	LLM Tokens ↓	LLM Ratio (%) ↓	Acc (%) ↑	Average Tokens ↓	LLM Tokens ↓	LLM Ratio (%) ↓
GSM8K								
Pure SLM	83.78	2190.7	0.0	0.0	83.78	2190.7	0.0	0.0
Pure LLM	90.86	1406.1	1406.1	100.0	94.62	1101.1	1101.1	100.0
Uniform Routing	87.58	1599.7	939.0	58.7	92.87	1222.5	762.8	62.4
CITER	86.73	1588.7	1129.8	71.1	93.26	1149.9	723.4	62.9
$\text{RouteLLM}_{\text{BERT}}$	88.85	1788.7	1256.6	70.2	88.38	1360.9	767.2	56.4
$\text{RouteLLM}_{\text{CASUAL}}$	87.52	1743.5	1062.4	60.9	91.82	1443.2	896.2	62.1
TokenRouter	90.07	1515.3	913.7	60.3	94.09	1070.6	635.9	59.4
MATH500								
Pure SLM	82.60	4500.0	0.0	0.0	82.60	4500.0	0.0	0.0
Pure LLM	88.40	3527.1	3527.1	100.0	89.00	2893.4	2893.4	100.0
Uniform Routing	84.20	3556.7	2041.5	57.4	85.40	4214.4	2414.9	57.3
CITER	84.40	4051.1	2536.7	62.6	86.20	4127.2	2539.0	61.5
$\text{RouteLLM}_{\text{BERT}}$	86.20	4208.4	2804.3	66.6	85.80	4036.2	2368.9	58.7
$\text{RouteLLM}_{\text{CASUAL}}$	84.80	4012.2	2616.1	65.2	86.80	4107.5	2850.9	69.4
TokenRouter	88.00	3880.9	2545.8	65.6	88.40	3974.0	2233.4	56.2
AIME24								
Pure SLM	26.67	11503.3	0.0	0.0	26.67	11503.3	0.0	0.0
Pure LLM	43.33	6811.2	6811.2	100.0	46.67	6363.5	6363.5	100.0
Uniform Routing	30.00	7253.2	4968.4	68.5	36.67	7067.8	5534.1	78.3
CITER	33.33	7372.4	5505.3	74.7	36.67	7774.7	5843.4	75.1
$\text{RouteLLM}_{\text{BERT}}$	36.67	8803.4	6097.8	69.3	40.00	8825.0	5885.4	66.7
$\text{RouteLLM}_{\text{CASUAL}}$	36.67	9023.7	6245.2	69.2	40.00	9061.9	6414.7	70.8
TokenRouter	43.33	6874.3	4981.8	72.4	46.67	6513.6	4948.3	76.0

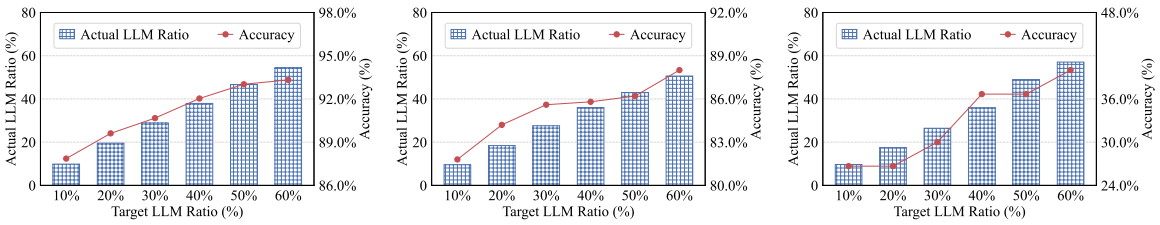
Evaluation Metrics. We assess all methods across two primary dimensions: overall performance and computational efficiency. Performance is measured by the final task accuracy. To quantify efficiency, we use average inference tokens as a hardware-agnostic indicator of computation and report the LLM token ratio (%), which denotes the proportion of tokens generated by the large model during inference. For self-hosted deployment, Appendix C additionally reports wall-clock latency. The LLM token ratio also remains a useful cost indicator in usage-based deployment settings.

4.2 Main Results

Table 1 compares TokenRouter with baseline methods across benchmarks. TokenRouter maintains near-LLM accuracy while substantially reducing large-model usage and overall computational cost. On the challenging AIME24 benchmark, TokenRouter matches the pure 14B LLM at 46.67% accuracy while reducing LLM token usage by 24%. The same pattern appears on GSM8K and MATH500, where TokenRouter recovers over 99% of pure-LLM accuracy (94.09% vs. 94.62% on GSM8K, and 88.40% vs. 89.00% on MATH500) while activating the LLM for under 60% of tokens. Compared with CITER and query-level RouteLLM, TokenRouter achieves higher accuracy under similar or lower budgets across all datasets, suggesting more effective capacity allocation when reasoning difficulty changes during generation. On GSM8K under self-hosted deployment, this reduction in LLM usage also yields lower wall-clock latency, as detailed in Appendix C. Overall, the results show that TokenRouter preserves reasoning quality while substantially reducing large-model usage.

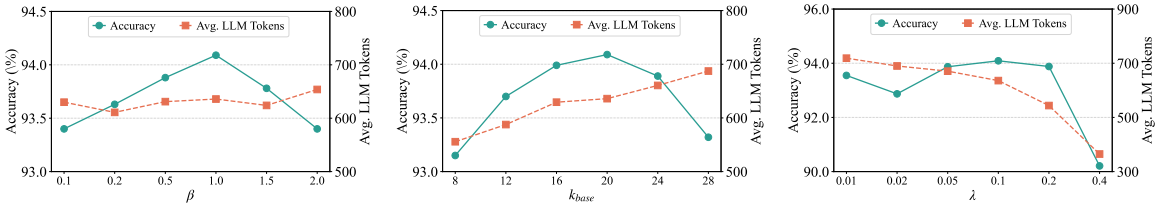
4.3 Budget-Controlled Trade-off

Figure 2 illustrates the trade-off between accuracy and efficiency under TokenRouter's budget control across the three datasets. The actual LLM usage ratio closely tracks the target ratio, showing that the self-calibrating policy enforces the specified budget reliably. Accuracy then improves smoothly as more budget is allocated. On GSM8K, increasing the LLM budget from 10% to 60% raises accuracy



(a) GSM8K (b) MATH500 (c) AIME24

Figure 2: TokenRouter's accuracy and actual LLM usage ratio under different target ratio budgets.



(a) Gain Factor β (b) Commitment Floor k_{base} (c) Smoothing Factor λ

Figure 3: Effect of varying key hyperparameters on TokenRouter's final accuracy and average LLM token usage.

from 87.86% to 93.32%. On MATH500, accuracy rises from 81.80% to 88.00% as the LLM activation ratio grows, and AIME24 shows the same monotonic trend. The resulting curves are smooth rather than erratic, suggesting that the controller adjusts capacity in a stable way as the target budget changes. These results indicate that TokenRouter lets users move predictably along the accuracy-cost curve, which is important for deployment under varying resource constraints. This behavior also indicates that budget tracking remains calibrated across problem difficulties.

4.4 Qualitative Analysis

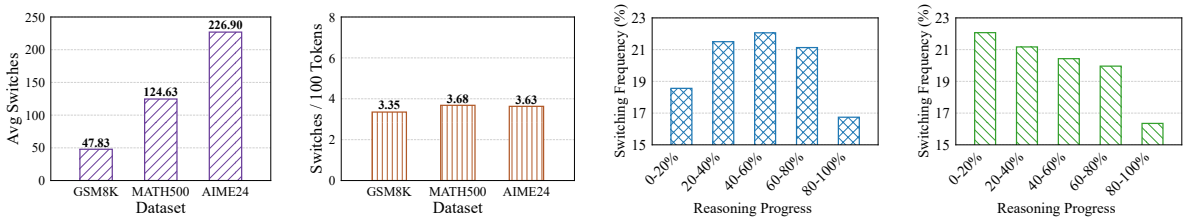
To better understand TokenRouter's decisions, we analyze representative moments when the controller activates the LLM. Table 2 shows four common scenarios: self-correction ("Wait, I made a mistake..."), planning under uncertainty ("Hmm, what is the next step?..."), pivotal logical deduction ("Therefore..."), and strategy adjustment ("But the question asks for..."). These examples suggest that the entropy-driven detector captures diverse high-difficulty reasoning moments rather than a single failure mode. Across all cases, entropy spikes align with conceptual transitions where continued SLM decoding may cause reasoning drift or premature conclusions. The triggers are therefore semantically meaningful rather than arbitrary local fluctuations. Selectively escalating to the LLM at these points helps preserve coherence while avoiding unnecessary computation. This pattern is consistent with using entropy as an online proxy for local reasoning difficulty.

Table 2: Examples of critical reasoning moments identified by TokenRouter. The table highlights distinct scenarios that need an LLM intervention to ensure reasoning integrity.

Reasoning Step	Entropy	S_t
Case 1: Self-Correction ...change is 20 - 17.25 = 2.75. **Wait**, I made a mistake...	3.21	4.87
Case 2: Uncertainty & Planning ...So $A = \pi \times 5^2 = 25\pi$. **Hmm**, what is the next step?...	4.11	5.23
Case 3: Logical Deduction ...The total distance is 300 miles. **Therefore**, the average speed is...	4.53	5.98
Case 4: Strategy Shift ...the fraction is $30/90 = 1/3$. **But** the question asks for the percentage...	2.98	4.12

4.5 Parameter Sensitivity Analysis

We further analyze the sensitivity of TokenRouter to its key hyperparameters on GSM8K, with results shown in Figure 3. Overall, the framework remains robust across a broad range of settings. Varying the gain factor β yields stable, non-oscillatory control, with the best balance between responsiveness and stability around $\beta = 1.0$. Adjusting the commitment floor k_{base} trades coherence against cost:



(a) Switch count (b) Switches / 100 tokens (c) GSM8K distribution (d) MATH500 distribution

Figure 4: TokenRouter's switching behavior across datasets.

Table 3: Ablation study of TokenRouter's core components.

Configuration	GSM8K		MATH500	
	Acc (%) ↑	Ratio (%) ↓	Acc (%) ↑	Ratio (%) ↓
w/o Robust Perception	93.04	67.1	86.60	62.9
w/o Budget Control	91.92	39.5	85.20	37.9
w/o Adaptive Commit.	87.86	10.9	83.20	9.7
w/o Commitment Floor	90.00	20.5	84.80	18.4
TokenRouter	**94.09**	**59.4**	**88.40**	**56.2**

smaller values can fragment reasoning, while larger ones increase computation, with $k_{base} = 20$ giving the best trade-off. The framework is somewhat more sensitive to the smoothing factor λ: very small values overreact to token noise, whereas large values respond too slowly to real complexity shifts. A setting around $\lambda = 0.1$ performs best. Taken together, these trends indicate that the method behaves predictably under moderate parameter variation and does not require careful tuning.

4.6 Ablation Study

We conduct an ablation study to evaluate the contribution of each core component in TokenRouter, with results summarized in Table 3. All modules contribute to the final balance between accuracy and efficiency. The adaptive commitment mechanism is the most critical: removing it drops GSM8K accuracy from 94.09% to 87.86%, indicating that fragmented intervention harms multi-step reasoning. Removing budget control also lowers accuracy by misallocating LLM usage, while removing robust perception makes the system more vulnerable to entropy noise, increasing the LLM usage ratio on MATH500 by nearly 7% while reducing accuracy. The ablations therefore align closely with the design motivation of the method: stable perception, budget regulation, and sustained intervention each address a distinct failure mode. Together, these results show that perception, control, and commitment play complementary roles in reliable fine-grained routing.

4.7 Analysis of Switching Behavior

To examine runtime behavior, we analyze TokenRouter's switching dynamics across datasets. Figure 4 shows that TokenRouter makes a moderate and consistent number of switches, typically fewer than five per problem. This indicates that the controller avoids oscillation and instead performs chunk-level routing aligned with reasoning structure. The switching distributions further show that most interventions occur in the middle stages of reasoning, where entropy rises and local reasoning difficulty is higher. Later arithmetic or summary stages trigger fewer LLM activations. This pattern indicates that TokenRouter deploys the large model selectively to difficult spans rather than switching indiscriminately, which is consistent with the intended role of the commitment mechanism. The moderate switching frequency also aligns with the lightweight routing overhead reported in Appendix C, further supporting a stable span-level routing pattern.

5 Conclusion

This work introduces TokenRouter, an adaptive runtime framework for token-level model routing. It combines a robust entropy detector, a self-calibrating control policy, and a dynamic commitment mechanism for stable intervention. Together, these components enable fine-grained allocation between large and small models. Our experiments show that TokenRouter retains over 99% of pure-LLM reasoning accuracy while activating the LLM for only 56.2% to 76.0% of generated tokens. These results support uncertainty-guided token-level routing as a practical mechanism for small-large collaboration in multi-step reasoning.

Limitations

Our study has two main limitations. First, we evaluate TokenRouter in a controlled paired-model setting for reasoning-oriented text generation, mainly on benchmark reasoning tasks and on small-large pairs from a single model family. This focus is useful for isolating token-level routing behavior and for analyzing the roles of uncertainty, budget control, and switching stability, but broader evaluation across additional task types, model families, and larger model suites would better characterize transfer beyond the current setting. Second, TokenRouter is intentionally built as a lightweight black-box controller that uses only decoding-time next-token probabilities and an entropy-derived uncertainty signal. This keeps the framework simple and architecture-agnostic, but the current study does not examine richer internal or external signals, nor more complex collaboration patterns such as larger routing suites or hierarchical coordination. We view these directions as natural extensions of the same routing perspective.

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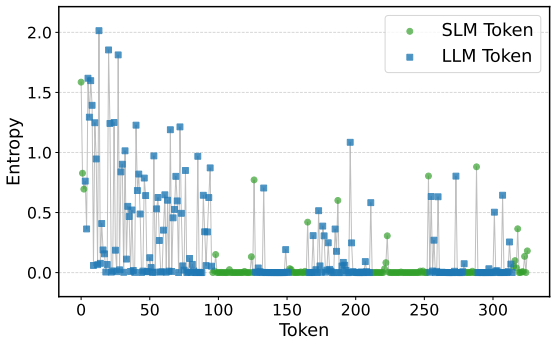
A Reasoning Dynamics Analysis

Figure 5 provides a detailed examination of the reasoning dynamics, revealing how TokenRouter adaptively allocates computational resources throughout the reasoning process.

As illustrated in Figure 5a, the entropy trajectory exhibits structured fluctuations that correlate with local reasoning difficulty. Sharp entropy spikes emerge at moments of uncertainty, signaling transitions into high-complexity reasoning phases that trigger LLM activation. In contrast, low-entropy segments correspond to routine narrative transitions, which the SLM handles efficiently.

Figure 5b aligns these entropy variations with the actual chain-of-thought trace. The LLM is predominantly engaged during conceptual abstraction, intermediate verification, or strategic adjustment, while the SLM dominates procedural computation and connective text generation. This alternation pattern shows that TokenRouter allocates additional capacity at spans where uncertainty and reasoning difficulty intensify. Overall, this analysis highlights the interpretability of TokenRouter's entropy-driven control. By synchronizing token-level routing with the evolving reasoning trace, the framework preserves coherence, avoids redundant computation, and achieves efficient yet faithful problem solving across complex reasoning trajectories with stable intervention patterns throughout the trace.



(a) Entropy trajectory where peaks mark complex reasoning moments.

Chain-of-thought reasoning process

To solve this problem, you must take into account:\n\n1. Calculating the total number of pages he needs to read.\n\n2. Determining the number of pages he has already read.\n\n3. Finding the remaining number of pages he needs to read.\n\n4. Calculating the average number of pages he must read each day for the remaining days </think>\n\n**Solution:**\n\n1. **Calculate the total number of pages Solo needs to read:**\n\n - Science: 4 pages\n - Social Studies: 20 pages\n - History: 7 pages\n - Geography: 8 pages\n\n \[\n \text{Total pages} = 4 + 20 + 7 + 8 = 39 \text{ pages}\n \]\n\n2. **Determine the number of pages Solo has already read:**\n\n - Pages read on Monday: 15 pages\n\n3. **Find the remaining number of pages to read:**\n\n \[\n \text{Remaining pages} = \text{Total pages} - \text{Pages read} = 39 - 15 = 24 \text{ pages}\n \]\n\n4. **Calculate the average number of pages to read each day for the remaining 4 days:**\n\n \[\n \text{Average per day} = \frac{\text{Remaining pages}}{\text{Remaining days}} = \frac{24}{4} = 6 \text{ pages/day}\n \]\n\n **Final Answer:**\n\n\[\n\boxed{6}\n\]

(b) Reasoning trace showing LLM engagement aligned with key reasoning shifts.

Figure 5: Reasoning dynamics and entropy-guided token routing. TokenRouter activates the LLM only at difficult reasoning steps, selectively allocating computational capacity to preserve reasoning coherence while maintaining high accuracy.

B CommonSenseQA Results

We further evaluate TokenRouter on the CommonSenseQA (Talmor et al., 2018) dataset to examine its generalization to broader reasoning domains. As shown in Table 4, TokenRouter achieves 62.6% accuracy under the 1.5B–7B configuration, closely matching the pure LLM baseline of 63.0% while activating the large model for only about 79% of the tokens. This corresponds to maintaining over 99% of the LLM's accuracy with a substantially reduced computational footprint. The results indicate that our entropy-driven routing strategy remains effective on commonsense reasoning, demonstrating consistent efficiency gains and robust adaptability across diverse task types. This consistency also suggests that the uncertainty signal continues to identify useful intervention spans beyond the main benchmarks in the paper.

Table 4: Results on the CommonSenseQA dataset. TokenRouter maintains over 99% of pure LLM accuracy while reducing large-model token usage by more than 28%.

Method	**Combination: 1.5B & 7B**			
	Acc (%) ↑	Average Tokens ↓	LLM Tokens ↓	LLM Ratio (%) ↓
Pure SLM	40.70	1222.8	0.0	0.0
Pure LLM	63.00	1013.6	1013.6	100.0
TokenRouter	62.60	922.4	721.8	78.3

C System Efficiency Analysis

Table 5 reports end-to-end wall-clock latency on GSM8K for a self-hosted deployment setting. TokenRouter reduces total latency from 30.335s to 19.028s on NVIDIA A100 and from 38.021s to 28.881s on NVIDIA RTX 4090. On NVIDIA H100, TokenRouter reduces total latency from 17.235s to 15.154s, corresponding to a 12.1% improvement. These results indicate that lower LLM

Table 5: End-to-end wall-clock latency on GSM8K in a self-hosted deployment setting. "LLM Time" and "SLM Time" denote the latency contributions of the two models within TokenRouter.

Hardware	Pure LLM Total (s)	TokenRouter Total (s)	LLM Time (s)	SLM Time (s)	Reported Improvement
NVIDIA A100	30.335	19.028	15.613	3.415	37.2%
NVIDIA RTX 4090	38.021	28.881	24.318	4.563	24.0%
NVIDIA H100	17.235	15.154	13.475	1.679	12.1%

usage can also translate into lower wall-clock latency when both models are locally hosted.

The routing decision logic is lightweight and runs on CPU. Across these runs, the average decision latency is between 6.253ms and 22.003ms. In addition, the implementation retains the KV cache for both the SLM and the LLM in GPU memory. When the controller switches back to a previously active model, the cached key-value pairs are reused. This reduces repeated context recomputation and helps keep routing overhead small relative to token generation cost.

TokenRouter introduces no additional GPU memory consumption for entropy computation because entropy is computed directly from the existing output logits. The additional CPU memory required for the controller state is less than 10MB.